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Review

Research progress in bird sounds recognition based on acoustic monitoring technology: A systematic review

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A B S T R A C T

Bird sound contains rich ecological information, and its related research results can be applied to animal behavior analysis, natural information collection and ecological environment monitoring. Since the early manual monitoring methods, many researchers have continuously innovated and improved the bird sounds recognition technology to overcome the long-standing drawbacks of long cycle time, high cost and poor effectiveness. These developments make bird sounds recognition a highly interesting, but also highly challenging research topic. Acoustic monitoring technology plays a vital role in the automatic recognition of bird sounds. With the popularization of acoustic monitoring technology, the technical routes based on traditional recognition models and neural networks have increased sharply, which has greatly promoted the development of bird sounds recognition. In view of these main technical routes, this paper summarized the research status of bird sounds recognition, provided a summary table of a variety of bird sound sample datasets, introduced the application evolution of various recognition technologies, and analyzed its open challenges. Meanwhile, this paper also cited the published experimental exploration on the improvement of deep learning networks. In general, this paper gives a comprehensive overview of the research process of bird sounds recognition based on acoustic monitoring technology, which has important theoretical and practical value to promote the development of bird sounds recognition technology, and provides a valuable reference for future related research.

1. Introduction

Birds are an important part of the ecosystem and play an integral role in maintaining the balance and stability of the whole ecosystem [1]. The species and number of birds predict habitat conditions and environmental quality changes [2]. The investigation and monitoring of birds can provide necessary information such as the species, quantity and quality of life of birds, help researchers grasp the status of bird resources, and provide a basis for effective protection, sustainable use and scientific management of bird resources [3,4]. But over the years, the investigation and monitoring of birds have faced challenges such as the complexity of the environment, the particularity and difference in bird habits [5].

In the investigation and monitoring of birds, the identification of bird species has always been a difficult problem for scientists [6]. The previous methods of bird species recognition mainly include manual survey and infrared camera monitoring, which requires the recognizer to master a lot of bird knowledge, and only apply to the situation with obvious morphological and behavioral characteristics [7–9]. These methods not only have the limitations of large labor and low efficiency, but also hurt the normal activities of birds [10].

As an important biological feature of birds, bird sounds have a high degree of recognition and have been widely used in bird species recognition research [11]. With the development of bird acoustics, sound-based bird recognition methods have gradually replaced those based on morphological or behavioral characteristics.

Bird sounds recognition is comprised of three fundamental components: preprocessing, feature extraction, and classification [12]. The analysis workflow of Bird sounds recognition is shown in Fig. 1. To better explain this process, the steps of this workflow are shown below:

Step 1: Audio preprocessing step such as resampling and denoising, is applied to determine valuable signal units, which can facilitate the subsequent stages.

Step 2: Audio feature extraction is carried out to transform the raw signal to representative features that retain the relevant information and discard the variability from unknown sources.

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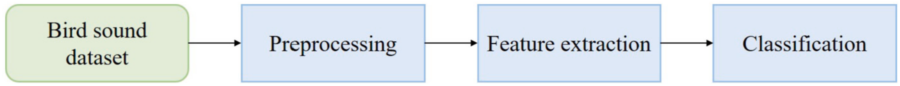

Fig. 1. A workflow summarizes the process of bird sounds recognition.

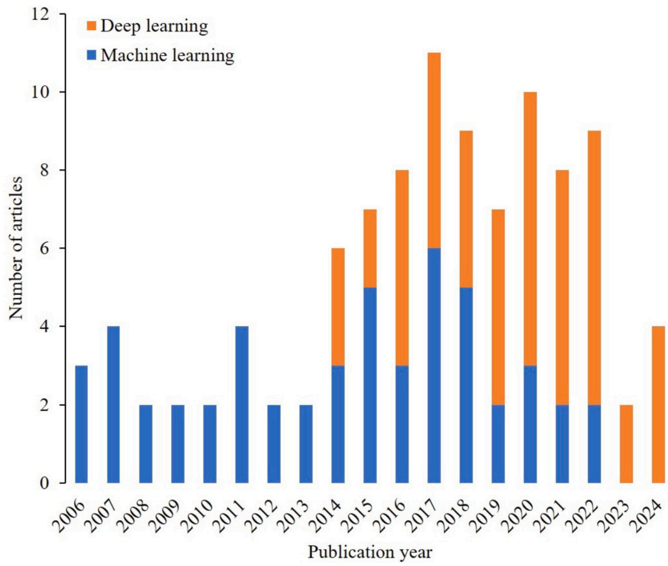

Fig. 2. The number of articles published on machine learning and deep learning in bird sounds recognition based on acoustic monitoring technology research (2006-2024).

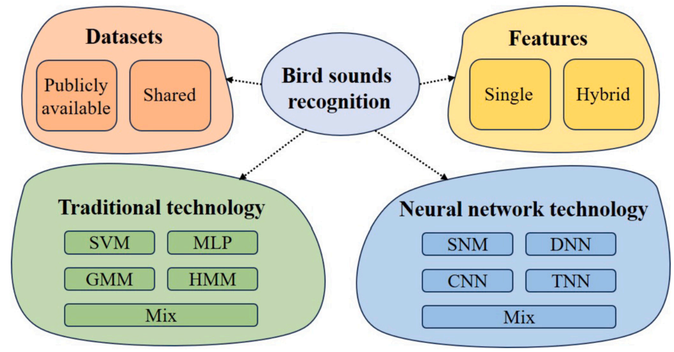

Fig. 3. Overview of bird sounds recognition in this paper. Abbreviations include Support Vector Machine (SVM), Multilayer Perceptron (MLP), Gaussian Mixture Model (GMM), Hidden Markov Model (HMM), Shallow Network Model (SNM), Deep Neural Network (DNN), Convolutional Neural Network (CNN), Temporal Neural Network (TNN).

Step 3: For the species classification step, classifiers map the extracted feature vectors to the related species and make decisions based on the precomputed models.

Automatic bird sounds recognition is an important technology to support bird monitoring and conservation activities, which can help researchers to automatically monitor the numbers and survival of specific bird species, environmental indicator species and endangered species [13]. Bird sounds recognition based on acoustic monitoring technology can speed up and improve the process of species recognition, and has strong scalability [14]. As early as the 1970s, there were records of birds using sounds to recognize relatives, and through recording and playback, it was found that birds can distinguish between the sounds of relatives and non-related individuals [15]. Meanwhile, with the rapid development of sound recording technology and digital technology, the methods of bird sounds recognition are increasingly strengthened, which promotes the development of bird sounds recognition technology [16,17]. At present, bird sounds recognition technology is mainly based on machine learning or deep learning and the use of both is shown in Fig. 2.

Researches into the use of acoustic monitoring technology in bird sounds recognition began with early acoustic models, primarily utilizing traditional machine learning methods for pattern recognition. An average rate of 2.75 articles per year in the 2006s (from 2 to 4 per year), 3.4 articles per year in the 2010s (from 2 to 6 per year), and 1.4 articles per year in the 2020s (from 0 to 3 per year). During this period, due to the rise of neural networks, deep learning-based methods grew significantly after 2014, and their usage has increased from 2 to 7 articles per year, gradually becoming mainstream.

With more and more scholars participating in the research of bird sound, the research of bird sounds recognition based on acoustic monitoring technology will continue to deepen and innovate. By surveying relevant literatures and integrating the latest review [18], this paper provides a comprehensive overview of bird sounds recognition Fig. 3, including dataset resources, features used, traditional technology and neural network technology. Meanwhile, the current research status and future research directions are also prospected.

The rest of this paper is organized as follows: in Section 2, several valid bird sound datasets are provided. In Section 3, different traditional bird sounds recognition methods are analyzed, including Support Vector Machines (SVM), Multilayer Perceptron (MLP), Gaussian Mixture Model (GMM), and Hidden Markov Model (HMM). In Section 4, deep learning-based bird sounds recognition methods are analyzed, and different research efforts are compared from shallow neural network models to deep neural network models. In Section 5, the investigation is summarized and the future research direction is prospected.

2. Data

In the research of bird sounds recognition, the quality of the dataset used is a critical factor in improving the recognition performance. By reviewing the existing literature in the field of bird sounds recognition, this paper collates the valid datasets of bird sounds provided in this literature. Table 1 provides detailed information about these datasets, including the introduction of each dataset, the species of bird sound, the number of samples, and the access to datasets.

There are 23 datasets in Table 1, the species of bird sound have been expanded from 2 to 1500, and the number of samples has also been expanded from 90 to 164210.

3. Traditional bird sounds recognition technology

More than a decade ago, many efforts were made in bird sounds recognition, and some traditional recognition techniques were proposed. These recognition techniques are based on machine learning, and mainly include template-matching classification methods and feature-based classification methods. Among the classification methods based on template matching, the most representative is the Dynamic time warping (DTW) algorithm. Although this method has high recognition accuracy, it is too computatively heavy, which seriously affects the recognition efficiency, and is not suitable for the task of bird sounds recognition. Feature-based classification methods are to establish a classification model by manually extracting features to realize bird sounds recognition, which is the most commonly used way.

Table 1
Datasets for bird sounds recognition.

Literature	Datasets	Species of bird sound	Number of samples	Access
[19]	LifeCLEF2016	999	34128	https://www.imageclef.org/LifeCLEF2016
[20]	LifeCLEF2017	1500	36492	https://www.imageclef.org/LifeCLEF2017
[21]	Xeno-Canto	659	50000	https://xeno-canto.org
[22]	DCASE2018	2 (absence /presence)	20000	https://dcase.community/challenge2018/task-bird-audio-detection-results
[23]	CPDO	14	1278	https://drive.google.com/open?id=1jE2JDcq35cWRnsMoPWE6bfqhDC4Vm9TE
[24]	ARBIMON	24	97900	https://arbimon.rfcx.org/
[25]	Zenodo	7	164210	https://zenodo.org/record/3338550
[26]	CLO-43DS	43	5428	https://datadryad.org/stash/dataset/doi:10.5061/dryad.j2t92
[27]	The assembled dataset (eBird, Xeno-canto, Macaulay)	984	22960	–
[28]	BirdCLEF subset	30	3000-30000	http://humbug.ac.uk/kiskin2018/
[13]	Birdsdata	20	14311	https://www.aminer.cn/
[10]	INECOL	7	301	http://www1.inecol.edu.mx/sonidos/menu.htm
[29]	NIPS4B	87	687	http://sabiod.lis-lab.fr/nips4b/challenge1.html
[30]	MLSP 2013	19	645	https://www.kaggle.com/c/mlsp2013-birds/data
[31]	Chiffchaff recording	13	41860	–
[32]	PELZ BIOPHON	18	7500	https://www.birdlife.cz/
[33]	COBRA	2	90	http://cobra.ic.ufmt.br
[34]	Bird recordings from Finland	8	3132	–
[1]	Treehugger	8	67000	https://www.treehugger.com
[35]	BIRDS	18	781	http://web.firat.edu.tr/turkertuncer/BIRDS.rar
[36]	Freesound	20	400	http://www.freesound.org
[37]	Lishui-Zhejiang Birdsdata, BirdCLEF2018-Small, BirdCLEF2018-Large	100, 150, 500	308, 1005 2513	https://github.com/WanderQY/ Hierarchical-taxonomy-aware-model
[38]	CBC2020	100	15032	https://xeno-canto.org

3.1. Feature-based classification methods

Feature-based classification methods mainly include Support Vector Machines (SVM) [39], Multilayer Perceptron (MLP) [40], Gaussian Mixture Model (GMM) [41], and Hidden Markov Model (HMM) [42].

3.1.1. Support vector machines (SVM)

Support vector machine (SVM) [39] is a generalized linear classifier for binary classification, with many variations, and uses variable parameters and kernels to solve the maximum margin hyperplane for learning samples. SVM was proposed in 1964, developed rapidly in the 1990s, and derived a series of improved and extended algorithms, which have been widely used in pattern recognition problems such as bird sounds classification [43].

Fagerlund [44] used two different parameter representations (the mel-cepstrum parameters and a set of low-level signal parameters) to study the method of automatic identification of bird sounds. In the decision tree, the SVM classifier for each node performing the classification between the two species was identified with 85% accuracy. Miguel et al. [45] compared the ability of three machine learning algorithms (SVM, linear discriminant analysis, and decision tree) to automatically classify sounds from 9 frog species and 3 bird species. Sounds was characterized by 11 variables in four standard parameters (minimum frequency, maximum frequency, call duration, and maximum power) with 94.95% accuracy in SVM. Marini et al. [46] expanded the number of bird species based on the manual extraction of MFCC features of bird sounds. The SVM classifier was used to identify the sounds of 50 species of birds, and obtained an accuracy of 45.9%. Although it did not obtain the ideal identification results, it provided valuable experience for multi-bird recognition. Lucio and Costa [47] proposed a different bird species identification method that used spectrograms generated by bird sounds to extract three texture descriptors (LBP, LPQ, and Gabor Filter). In their study, 46 species were collected from the Xeno Canto database and the optimal recognition accuracy of the SVM classifier was 77.65%. Zottesso et al. [48] extended the work described in [47] by automatically splitting the input signal. In their work, 45 species were collected from the Xeno Canto database, and an identical image texture descriptor (Local Binary Pattern, LBP) was used to characterize bird sounds. The experimental results showed that the best recognition accuracy of SVM classifier was 78.97%, which had a slight improvement. Jimmy et al. [49] proposed a new method to improve the performance of the classifier, which used short-time feature extraction and time feature integration. A non-negative matrix decomposition was performed on the spectrogram of 10 bird species to characterize bird sounds, and SVM was used for the final classification with 76.4% mean average precision (MAP). Zottesso et al. [50] collected 915 species of bird sounds samples from the competition database of the LifeClef 2015, and divided the number of identified species into a range of 23 to 915 by the division of subset recordings. The SVM classifier was equipped with an RBF core with tradeoff parameters, and achieved the recognition accuracy of 71%. Ramashini et al. [51] extracted three types of cepstrum features and used SVM to classify them separately. On 15 endemic Bornean bird sounds, models using Gammatone frequency cepstral coefficients (GTCC) outperformed models using MFCC or Linear prediction cepstrum coefficients (LPCC) features with 93.3% accuracy.

3.1.2. Multilayer perceptron (MLP)

Multilayer perceptron (MLP) [40] is a forward structure that maps a set of input vectors to a set of output vectors, where each node is a neuron with a nonlinear activation function. MLP is a generalization of perceptron, which overcomes the shortcoming that perceptron cannot recognize linear unfractional data. Meanwhile, MLP was a very popular machine learning method in the 1980s, and had a broad application prospect in bird sounds recognition [1].

Selin et al. [34] used wavelet analysis to extract features, which retained frequency and time information in audio. The feature vectors calculated by wavelet coefficients are used as input to MLP and Self-Organizing Map (SOM) classifiers, and tested on recordings of 8 bird species. MLP (96%) had higher identification accuracy compared to SOM (78%). Cai et al. [52] studied the performance of different pretreatment methods in bird species recognition, and effectively applied a noise reduction algorithm. Finally, the MLP was verified on recordings of 14 bird species, using MFCC as input feature, and achieved 86.8% accuracy. Lopes et al. [53] used MFCC feature set to evaluate machine learning algorithms (KNN, SVM, MLP). The comparison experiment was carried out in the recordings of 3 bird species, and the accuracy of the experimental results was 79.2%, indicating that the automatic bird species recognition problem based on sounds is very promising. Compared with the common classification of acoustic patterns in a single type of soundscape, Zhang et al. [54] focused more on the recognition of synchronous acoustic patterns in long-term bird sound recordings. In their work, the acoustic index was used as a global feature and the MLP was used as a base classifier. Through the verification of 5 kinds of bird sounds collected in real scenes, the experimental results showed that compared with single label classification, the multi-label classification method provided more detailed distribution information on various acoustic modes, and could effectively promote the investigation of biodiversity. Albornoz et al. [55] used the correlation principle of speech recognition and adopted a speech activity detection method to analyze the status of birds before feature extraction. Verified on a dataset of 25 bird species, MLP achieved a maximum accuracy of nearly 89.3%, speeding up and improving the bird identification process. Pahuja et al. [1] extracted the feature matrix based on the short-term Fourier transform spectrogram and used it to characterize the vocal patterns of birds after statistical evaluation. Using a dataset of eight common Eurasian birds, a multi-layer perceptron neural network (MLP-NN) classifier was trained to achieve improved recognition accuracy (96.1%), recall (82.6%), and precision (84.5%).

3.1.3. Gaussian mixture model (GMM)

Gaussian mixture model (GMM) [41] is to use the Gaussian probability density function to accurately quantify things, and decompose a thing into several models based on the Gaussian probability density function. Gaussian distribution can well describe the spatial distribution and characteristics of data in parameter space, and Gaussian density function has the advantage of easy parameter estimation [56]. Therefore, Gaussian mixture model is widely used in pattern recognition, such as the field of bird sounds recognition.

Lee et al. [57] calculated a feature set derived from the two-dimensional Mel frequency cepstrum coefficients to simulate different syllables of the same bird species. For each species, set a different amount of training data to find the most appropriate GMM group score. Then, the average vector of the GMM is used as the prototype vector for each bird species. In the experiment, 28 species of birds were classified, and the best classification accuracy was 84.06%. Cheng et al. [58] broke the deadlock between call dependence and species recognition in animal acoustic recognition research, and proposed a call-independent and automatic individual recognition method. In their experiment, MFCC was used as the input feature and GMM was used as the basic classification model. An accuracy of 89.1% to 92.5% was achieved for the sounds of four passerine birds. Jančovič and Köküer [59] first proposed a method to automatically detect and recognize bird sounds in noisy environments. Based on the spatiotemporal region of bird sounds, the tonal features are extracted from the sinusoidal components of the short-time spectrum. GMM was used as the standard model to identify 165 different bird syllables, which were produced by 95 species of birds. The recognition accuracy was 95.4%, showing the high accuracy of bird sounds detection. Lee et al. [15] changed the existing feature extraction operation, regarded the corresponding spectral graph as a gray-level image, and used the MPEG-7 angular radial transform (ART) descriptor to describe the shape features of image. The shape features of the image are used as a new feature descriptor to represent the bird sound fragment in a fixed duration. The GMM classifier was used to classify 28 species of birds, and the best classification accuracy of the proposed ART descriptor of bird sound fragments in the 3-second and 5-second was 86.30% and 94.62%, respectively, which was better than the commonly used descriptors. Deepika and Nagalinga [60] improved the ART descriptor in [15] and proposed a sector expansion algorithm. The spectrogram images obtained from bird sound clips of fixed duration were converted into sector images, and the GMM classifier was used to identify the bird species in the recordings. The best recognition accuracy was 98.6%, indicating that the new ART descriptor achieved better results. Ptacek et al. [31] adopted voice activity detection algorithm to automatically mask noise and unneed sounds, and proposed a bird individual recognition system based on GMM and Universal Background Model (UBM). With the verification in the dataset of 13 bird species, the overall recognition accuracy was 78.5%, indicating that the recognition system based on acoustic monitoring technology was feasible for processing the individual recognition of birds recorded in real environment.

3.1.4. Hidden Markov model (HMM)

Hidden markov model (HMM) [42] is a probabilistic statistical analysis model about time sequence, describing a process of generating a sequence of states from a hidden Markov chain, and then generating a sequence of observations from the sequence of states. HMM was proposed in the 1970s, spread and developed in the 1980s, and gradually became an important research direction of bird sounds recognition [56].

Chou et al. [61] segmented bird sounds into many syllables to obtain several primary frequency sequences. Fuzzy C-mean clustering method was used to cluster frequency sequences of all syllables and generate different syllable groups. Each syllable group was modeled based on HMM, and the recognition process was realized by finding the most matched HMM template. The proposed method was tested on 429 species of birds, and the recognition accuracy can reach more than 78%. Potamitis et al. [62] built a practical modular framework with HMM based on the computer, and applied it to the task of continuous detection of specific bird species in the field. Two large real-world databases were evaluated with identification accuracy of 71.2%-85.1% and recall of 77%-91.3%, facilitating further use of the Digital Autonomous Recording Units (ARUs) for species identification. Jančovič and Köküer [63] extended HMM from the single-species bird sound recognition problem to the multi-species problem, and proposed a multi-species bird sound recognition algorithm based on penalized maximum likelihood. In their study, HMM was used to model the time evolution in the frequency track of each bird sound. The number and species of birds were determined by possibility maximization detection. The experimental evaluation was carried out on the field recordings containing 30 species of birds, and the recognition accuracy was 78% when the number of species was known, and 69% when the number of species was unknown, which improved the recognition performance of multi-species birds. Ntalampiras [64] designed a transfer learning framework using HMM and Reservoir Network to transfer the experience of music genre classification to the bird sounds recognition task. With the feature space transformation operation, the species distribution among bird sounds were simulated by the probability density distribution of 10 different music types. The framework was tested on 10 European bird species and the recognition accuracy was 92.5%, with an average improvement of 11.2%. Stastny et al. [32] calculated the human factor cepstral coefficients (HFCC) from the given signal to characterize specific bird sound recordings. Then, voice activity detection was used to detect the activity of bird vocal fragments. For each bird detected, only one corresponding HMM was used to calculate the likelihood rate corresponding to the coded value of that species. With the test on 18 species of birds, the best accuracy was 90.45%. In order to reduce processing time and resource consumption, De Oliveira et al. [33] evaluated the effectiveness of three data reduction methods. These methods were applied after feature extraction to reduce the amount of computation. Experiments showed that the training time was shortened by 99.6%-99.7%, but the recognition performance was not significantly reduced when the number of types was maintained and the number of training samples was reduced by 10 times by using the MFCC input after data reduction into the HMM classifier.

3.2. Discussion of traditional technology

3.2.1. Comparative analysis of different models

Various machine learning models exhibited distinct performances in bird sounds recognition task. SVM demonstrated good classification performance in many researches because of its principle of maximum

Table 2
The advantages and disadvantages of different traditional recognition methods, including Support Vector Machines (SVM), Multilayer Perceptron (MLP), Gaussian Mixture Model (GMM), and Hidden Markov Model (HMM).

Methods	Advantages	Disadvantages
SVM	Simple algorithm and fast training speed.	Difficult to implement in large-scale training samples.
MLP	High parallelism and strong self-adaptation.	Difficulty in selecting the number of implicit nodes and slow speed of learning.
GMM	Easy parameter selection and good inclusion between categories.	Computationally heavy and not flexible enough.
HMM	Acoustic characteristics vary flexibly and good robustness.	Over-smooth acoustic features and fuzzy target features.

Table 3
The citation details of traditional recognition methods in bird sound recognition.

Year	Study	Number of classes	Feature use	Classifier	Performance
2007	Fagerlund [44]	2	Two different parametric representations	SVM	Accuracy: 85%
2009	Miguel et al. [45]	12	Four standard call variables	SVM	Accuracy: 94.95%
2015	Marini et al. [46]	50	MFCC	SVM	Accuracy: 45.9%
2015	Lucio and Costa [47]	46	Textural features	SVM	Accuracy: 77.65%
2016	Zottesso et al. [48]	45	Textural features	SVM	Accuracy: 78.9%
2017	Jimmy et al. [49]	10	Non-Negative Matrix Factorization	SVM	MAP: 76.4%
2018	Zottesso et al. [50]	915	Spectrograms	SVM	Accuracy: 71%
2022	Ramashini et al. [51]	15	Linear prediction cepstrum coefficients (LPCC), MFCC, Gammatone frequency cepstral coefficients (GTCC)	SVM	Accuracy: 93.3% (Highest)
2006	Selin et al. [34]	8	Four features derived from wavelet packet decomposition coefficients	MLP	Accuracy: 96%
2007	Cai et al. [52]	14	MFCC	MLP	Accuracy: 86.8%
2011	Lopes et al. [53]	3	MFCC	MLP	Accuracy: 79.2%
2016	Zhang et al. [54]	5	Acoustic indices	MLP	Accuracy: 85.3%
2017	Albornoz et al. [55]	25	Voice activity detection (VAD)	MLP	Accuracy: 89.3%
2021	Pahuja et al. [1]	8	Short Time Fourier Transform spectrogram	MLP-NN	Accuracy: 96.1%
2008	Lee et al. [57]	28	Two-dimensional MFCCs	GMM	Accuracy: 84.06%
2010	Cheng et al. [58]	4	MFCC	GMM	Accuracy: 89.1%–92.5%
2011	Jancovic and Köküer [59]	95	Tonal feature	GMM	Accuracy: 95.4%
2013	Lee et al. [15]	28	Image shape features	GMM	Accuracy: 86.3% (3s), 94.62% (5s)
2014	Deepika and Nagalinga [60]	-	Image shape features	GMM	Accuracy: 98.6%
2016	Ptacek et al. [31]	13	VAD	GMM+	MAP: 78.5%
2007	Chou et al. [61]	420	Syllable group	HMM+	Accuracy: 78.3%
2014	Potamitis et al. [62]	-	-	HMM	Recall: 77%–91.3%, Precision: 71.2%–85.1%
2015	Jančovič and Köküer [63]	30	Frequency track	HMM	Accuracy: 78% (known), 69% (unknown)
2018	Ntalampiras [64]	10	Feature space	HMM+	Accuracy: 92.5%
2018	Stastny et al. [32]	19	HFCC	HMM	Accuracy: 90.45%
2020	Oliveira et al. [33]	3	MFCC	HMM	Accuracy: 99.6%-99.7%
2017	Zhao et al. [65]	11	Spectrograms	GMM+SVM	F1-Score: 92.8%
2017	Liu et al. [66]	5	MFCC	GMM+SVM	Accuracy: 20% (improvement)
2018	Xu et al. [67]	11	Four acoustic features	SVM+RF	Accuracy: 92%

boundary hyperplane. MLP, owing to its nonlinear mapping capabilities, showed high accuracy in processing complex bird sound data. GMM had advantages in simulating the spatial distribution of data, particularly effective for sounds exhibiting Gaussian distribution characteristics. HMM standed out in multi-species bird sounds recognition due to its ability to handle time-series data.

With the use of sound datasets for various recognition tasks, Table 2 provides a brief overview of the advantages and disadvantages of SVM, MLP, GMM, HMM. SVM has simple algorithm and fast training speed, but it is difficult to implement for large-scale training samples. MLP has a high degree of parallel processing and very strong adaptability, but it is very difficult to select the number of hidden nodes in the network, and the learning speed is slow. The parameter selection of GMM is convenient, but the calculation is large and not flexible enough. The acoustic features of HMM are flexible and robust, but the acoustic features are too smooth, which makes the generated target features fuzzy, thus reducing the recognition effect.

In addition, audio signals and acoustic features used in most of the above methods were manually segmented. This is unfavorable for bird acoustic detection which needs to process a large number of soundscape recordings. Even though some studies proposed automated segmentation for feature extraction, it was still affected by the generalization ability of parametric features.

Considering these limitations, future research can enrich datasets by collecting more "unknown" acoustic events, enhance adaptability in complex environments using sound source separation techniques, and improve accuracy and robustness by integrating additional classification models.

3.2.2. Citation analysis of different models

Various manual acoustic features and traditional classification methods based on the development of machine learning have been constantly cited in the acoustic detection of bird sounds. The relevant literatures cited have been sorted out, and the details are shown in Table 3.

Table 3 provides a summary of the use of traditional bird identification techniques, including descriptive information of the literature, number of bird species, acoustic features used, classification methods, and performance results. SVM accuracy in bird sounds recognition

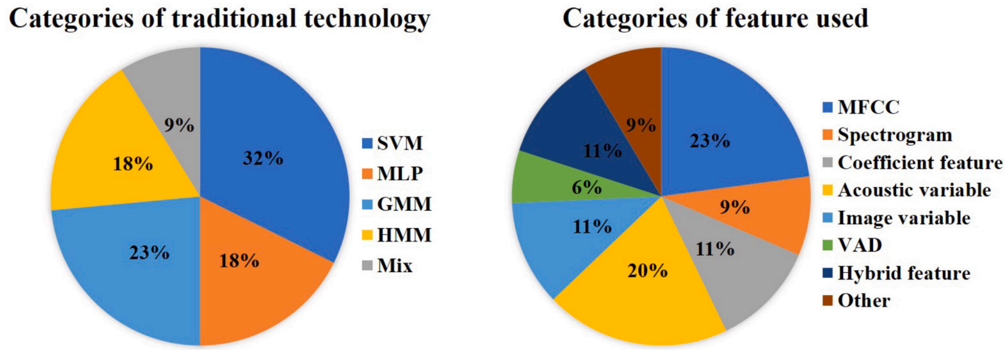

Fig. 4. Percentage pie chart of categories of traditional technology and feature used. Coefficient feature, Acoustic variable, Image variable are not specific feature names, but categorizations of some features.

ranged from 45.9% to 94.95%, demonstrating its applicability and effectiveness in different studies. MLP showed broad potential in bird sounds recognition, with accuracy ranging from 79.2% to 96%. GMM achieved high recognition accuracy ranging from 84.06% to 98.6%, through different training data volumes and feature sets. HMM accuracy in bird sounds recognition ranged from 69% to 95.4%, demonstrating its effectiveness in handling multi-species problems and noisy environments.

To more directly illustrate data distribution, this paper presents pie charts depicting the percentage of categories of traditional technology (Left in Fig. 4) and categories of feature used (Right in Fig. 4), respectively. In the categories of traditional technology, SVM, MLP, GMM, HMM and Mix accounted for 32%, 18%, 23%, 18%, 9%, respectively. This shows that SVM is the most frequently employed technique, while the mix, though least utilized, is still a significant component. In the categories of feature used, the proportion from most to least is MFCC (23%), Acoustic variable (20%), Coefficient feature (11%), Image variable (11%), Hybrid feature (11%), Spectrogram (9%), Other (9%), VAD (6%). As a common feature in speech and audio processing, MFCC is most frequently used in bird sound recognition based on traditional technology. Hybrid features occupy a certain proportion, which shows that it is a common practice to combine multiple features to improve model performance in feature engineering.

By integrating these research findings, bird sounds recognition requires a comprehensive consideration of feature extraction, model selection, data preprocessing, and environmental factors. Feature extraction is a critical step in bird sounds recognition, including MFCC, HFCC, etc. Combined with dimensionality reduction technology, it can reduce the computational load while maintaining the recognition performance. The performance of the selected model is a critical factor in practical applications, particularly for the more challenging task of multispecies identification, which requires more complex models and algorithms to distinguish avian acoustic characteristics. Data preprocessing methods, such as noise reduction and sound activity detection, also play a crucial role in improving model performance. The accuracy of bird sound recognition is susceptible to recording environment effects, modeling and suppression of environmental noise could enhance performance in natural settings.

4. Neural network technology for bird sounds recognition

In bird sounds recognition, the structures of sound signal are complex, and the simple models based on machine learning tend to have the problem of weak data representation ability. Therefore, it promotes the birth and rapid development of neural network technology. Based on deep learning, neural network technology can accept much larger dimensions of input data than traditional recognition technology, and can also automatically learn the features of the data, significantly shortening the extraction time of acoustic features [21]. According to the structure and number of network layers, the neural network technology for bird sounds recognition can be divided into shallow structure and deep structure.

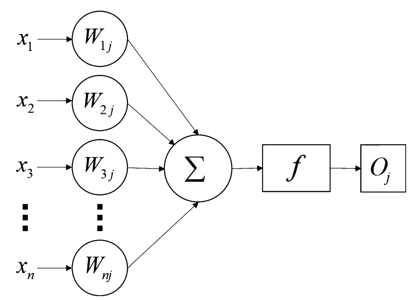

Fig. 5. The structure diagram of ANN simulated human brain neurons [68]. x_1, x_2, x_3, …, x_n represent the values sent to each input connection. W and f represent input function and activation function, respectively. O represents the output of the neuron.

4.1. Shallow network models

The shallow network models applied to bird sounds recognition are mainly composed of the basic Artificial Neural Network (ANN) and some of its deformation networks. As a core component of deep learning algorithms, the name and structure of ANN is inspired by the human brain, by using computers to mimic the way biological neurons transmit signals to each other [68,69]. Fig. 5 shows the basic structure of simulation of ANN for human brain neurons.

In Fig. 5, each neuron has input connections and output connections. Each input connection has a weight, which means that the values sent to each input connection x_1, x_2, x_3, …, x_n need to be multiplied by the corresponding weight factor W. Since there may be many input values entering a neuron, each neuron sets an input function $\sum$ to sum all the values. Typically, the input values of the joins are weighted and summed. The processed values are passed to the activation function f, which is used to calculate whether to send some signal to the output of this neuron.

The basic structure of ANN is mainly composed of node layers, as shown in Fig. 6. In the network structure diagram, the ANN consists of one input layer (pink), one or more hidden layers (blue), and one output layer (green), from left to right. Each node is called an artificial neuron, and connected to another node, with associated weights and thresholds. If the output value of any single node is higher than the specified threshold, the node is activated and the data is sent to the next layer of the network. Otherwise, the data will not be delivered.

By imitating the structure and function of the human brain and drawing on the research results of biological neuroscience, ANN realizes the information processing. As an emerging interdisciplinary subject, ANN not only promotes the application and development of intelligent computing, but also brings revolutionary changes to the research methods

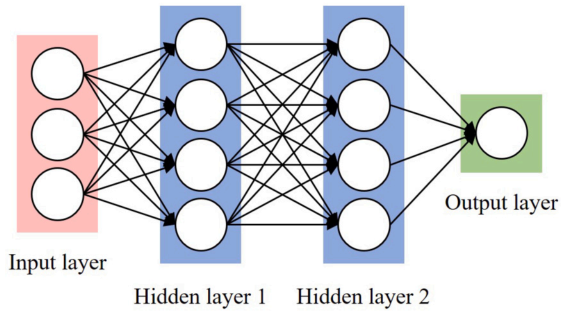

Fig. 6. The network structure diagram of ANN [69].

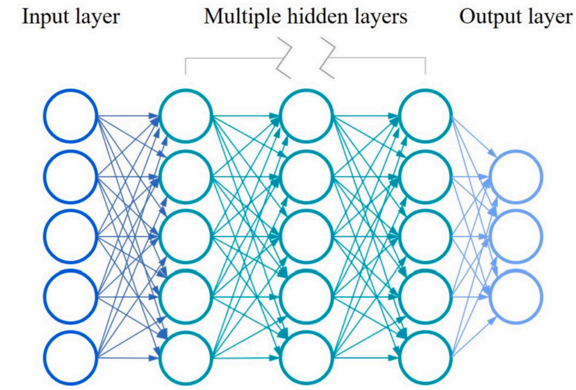

Fig. 7. Hierarchy of DNN [73].

of information science and neurobiology. At present, ANN has been successfully applied in bird sounds recognition.

Raghuram et al. [70] applied the three-layer ANN classifier to the bird sounds classification experiment and set up 24 neurons in a single hidden layer. To calculate the average frequency value of each bird sound recording, the fast Fourier transform was applied to extract specific tonal components of the sounds of bird. Finally, pitch values and energy values were used as the input of the classifier, and the classification accuracy of 93.93% was obtained on the dataset of 33 bird species. Mohanty et al. [71] upgraded Spiking Neural Network (SNN), a third-generation artificial neural network based on ANN, to improve the performance of bird species recognition system. The sound waves generated by birds were analyzed by Attribute extraction methods, and a Permutation Pair Frequency Matrix (PPFM) was obtained as the characteristics of bird sounds. The performance of SNN was evaluated on a dataset of 13 bird species and the accuracy was 92%. Mohanty et al. [23] conducted simulation verification of the method provided by [71]. Before preprocessing the original bird sounds, the noise was removed using discrete wavelet transform. Then, the syllable segmentation method provided in [71] was used to extract features from the spectrogram, and then the features were standardized to calculate PPFM. Finally, SNN was used to classify and recognize the sound data of 14 bird species, and the recognition effect was described in [71]. Ghani and Hallerberg [11] obtained sound datasets of 10 bird species from the Xeno-Canto database and randomly created ten different subsets using the bag-of-birds method. The ANN classifier was trained on six pre-calculated sound features (spectral centroid, spectral roll-off, zero-crossing rate, spectral bandwidth, root-mean-square energy, and MFCC), and the MAP achieved between 86% and 94%. The results showed that classification effect obtained by ANN on different subsets depended on the number of species, and the accuracy of ANN tended to decline with the increase in the number of species.

4.2. Deep network models

Deep network model is to build an architecture model with multiple hidden layers, and automatically extracts more representative features through large-scale data training. Different from the layer limit of shallow network model, the deep network model can choose any number of layers according to the requirements of designer [72]. Deep network model is not only widely used in image recognition, but also extended to the field of voice recognition, and has become a new focus of international research. Currently, in bird sounds recognition based on deep network models, Deep Neural Network (DNN) [73], Convolutional Neural Network (CNN) [74] and Temporal Neural Networks (TNN) [75,76] are gradually developed.

4.2.1. Deep neural network (DNN)

Hition [77] put forward the concept of deep learning, alleviated the problem of local optimal solution BY pre-training, and increased the hidden layer to 7 layers, achieving the true sense of “depth”. There was no fixed definition of “depth”, and a four-layer network could be considered “deep” in speech recognition, while a network of more than 20 layers was common in image recognition. Fig. 7 shows the classic DNN hierarchy [73] consisting of one input layer (blue), multiple hidden layers (green), and one output layer (purple) from left to right. The input layer, the hidden layer and the output layer constitute the three components of DNN and are fully connected (FC) with each other. In terms of structure alone, there is not much difference between the fully connected DNN and the ANN in the shallow networks. However, the “depth” in the deep neural network means that the hidden layer is greater than 2 layers, which gives it stronger ability of abstraction and dimensionality reduction.

Different from manual extraction of features or expert design rules, DNN can automatically obtain effective information for learning, and is the basis of artificial intelligence applications. Meanwhile, the breakthrough application in speech recognition has made the application of DNN in bird sounds recognition grow rapidly.

Koops et al. [78] used three hidden layers and dropout to build a DNN model with multiple topologies and identify the sound records of 501 species of birds. The experimental results showed that when the combination of MFCC, mean and variance of MFCC, delta-MFCC and delta-delta-MFCC was selected as inputs, the model had the best recognition effect, and could correctly classify 73% of bird audio clips. Boulmaiz et al. [79] detected the tonal regions in bird sounds containing noise, and extracted the gammatone teager energy cepstral coefficients (GTECC) features to represent the sound signal. On a dataset of 36 bird species from Tonga Lake (northeast of Algeria), the DNN model was used as the classifier, and the equal error rate was 18.47%. Fazeka et al. [80] further proposed a multi-modal bird sounds recognition method based on DNN model. The audio sample and metadata of bird sounds were two different data modes but were simultaneously used as inputs to the model. The audio samples were fed into a neural network model with four convolutional layers, the metadatas were fed into a fully connected layer, and the results of both were combined through a fully connected layer. Verified on a dataset of 1,500 bird species from the BirdCLEF 2017 mission, the method achieved a MAP of 51.1% to 57.9%. Chakraborty et al. [81] proposed a classifier based on dynamic kernel SVM and DNN to recognize bird sounds. In this work, each bird sound record was represented as a set of feature vectors of different lengths, and the recorded sound signal was characterized by MFCC features. The results showed that the DNN model had a good performance, with an accuracy of 63.1% to 91.5%. Jancovi et al. [82] developed an unsupervised modeling method of mixed HMM and DNN, to identify the vocalizations of a variety of birds in the given recordings. The recordings of the bird sounds were broken down into distinct segments, and each segment was further represented as a sequence of frequencies and amplitudes. In the process of establishing acoustic model, the input sequence was initialized based on HMM model and then classified by DNN model. The method was verified by the dataset of 48 bird species, and the recognition accuracy

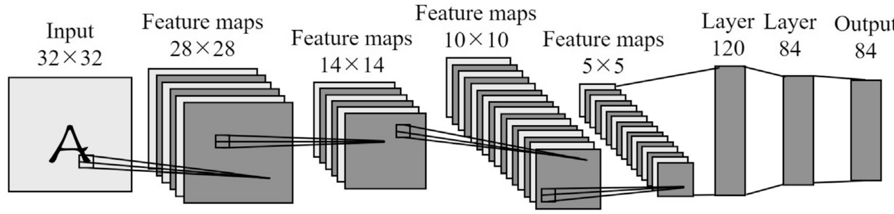

Fig. 8. The network structure of CNN [83].

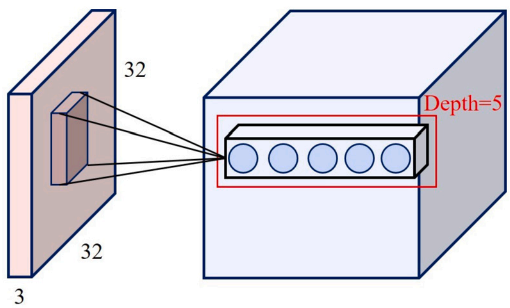

Fig. 9. The structure of the convolution layer [74].

was 95.4% to 98.7%. Kahl et al. [27] developed a DNN model called BirdNet and improved the classification performance of the model by inputting high temporal resolution of bird spectrograms. Through data preprocessing, enhancement, and mixing training, 984 North American and European bird species were identified by the BirdNet. The experimental results showed that the neural network model could effectively identify hundreds of bird vocalizations from a large amount of audio data, achieving the MAP of 79.1% on the single species recordings.

4.2.2. Convolutional neural network (CNN)

Convolutional Neural Network (CNN) [74], a class of feedforward neural networks with convolutional computation and deep structure, is one of the representative algorithms of deep learning. A typical CNN structure is mainly composed of three parts: convolutional layer, pooling layer and fully connected layer [83]. However, CNN is not a fixed three-layer structure, but a multi-layer structure, as shown in Fig. 8.

Fig. 8 shows a basic CNN hierarchy consisting of two “convolution layer - pooling layer” operations and a fully connected layer. Through representation learning, CNN can classify the input information according to its hierarchical structure. When the input data is transmitted in the network, CNN can continuously reduce the dimension of large data into small data, and effectively retain the essential characteristics of the input object. Finally, the essence of the input data is exported.

The convolutional layer [74] is the most important part of CNN, and its structure is shown in Fig. 9. In the given structure diagram, the left side is the input data of size $32 \times 32 \times 3$, and the right side is a convolution layer of depth 5. Unlike the traditional fully connected layer, the input of each node in the convolutional layer is only a small piece of the previous network layer.

CNN model can effectively mine the information of local structure, and has been developed rapidly in the field of bird sounds recognition [84].

Sprengel et al. [85] collected 999 species of birds in the dataset of BirdCLEF 2016 bird Recognition Competition, and extracted the spectrogram from the audio through a novel data preprocessing method. Under different prediction conditions, the average accuracy of 55.5%-68.6% was obtained by using CNN model. Tóth and Czeba [86] applied AlexNet, an improved deep convolutional neural network, to bird sounds recognition, and used a fixed-size spectrogram as input. In the BirdCLEF 2016 Bird Identification competition, the MAP for major species was more than 40%, and the MAP for a mix of major and background species was more than 33%. Kahl et al. [87] used a variety of CNN models to challenge the LifeCLEF2017 bird recognition task. Tested on audio recordings of 1,500 bird species, the method achieved the MAP of 60.5% to 68.7%. Fritzler et al. [88] converted bird audio recordings into spectrograms and used pre-trained Inception-v3 convolutional neural networks for bird sounds recognition. In the BirdCLEF 2017 task, the results of network were evaluated using audio recordings of 1,500 bird species, achieving the MAP of 49.6% to 56.7%. Sankupellay and Konovalov [89] used a deep convolutional neural network architecture, ResNet-50, to automatically identify bird sounds. The spectrograms were extracted from 46 species bird sounds as input to the model, and the recognition accuracy was 60%-72%. Stowell et al. [90] summarized machine learning techniques and used CNN model for classification on a new acoustic monitoring dataset. The experimental results showed that the method could obtain about 88% accuracy in remote monitoring of bird sound data. Incze et al. [91] used the bird audio dataset obtained from Xeno-Canto to generate spectrograms, and performed different configurations and hyperparameter adjustments on the pre-trained CNN model, MobileNet. In the course of testing, this method outperformed other recognition models and achieved an accuracy of 70.9%. Kucuktopcu et al. [14] adopted a multi-layer neural network algorithm to build a CNN model to recognize 21 species of bird sounds. As the number of network layers increased, the recognition accuracy of the model increased from 87% to 90%. Ruff et al. [25] used deep convolutional neural networks (DCNN) to automatically detect owl calls in real environments. On seven kinds of owl sound datas recorded in the field, the frequency spectrum of recordings was extracted as network input, and the recognition accuracy was up to 91.5%. Xie et al. [26] used the three time-frequency representations of bird sounds to train three CNN models respectively, and then carried out pairwise splicing and fusion. The fusion model was verified on 43 bird species data sets, and the optimal fusion model could reach 86.31% MAP and 93.31% F1-score. Tubaro and Mindlin [92] synthesized the simulation data of 6 species of bird sounds by using the dynamic system. In this work, a CNN model was established for bird sounds recognition, which achieved 89% accuracy, 84% recall and 84% F1-Score. LeBien et al. [24] trained the CNN model based on the Mel spectrum extracted from bird sounds, and used transfer learning to reduce the amount of data and time required. The recognition experiments on the dataset of birds and frogs achieved the MAP of 89.3%. Kiskin et al. [28] proposed a convolutional neural network structure based on wavelet transform, which used wavelet transform to transform input signals instead of STFT. The transformed features were used to train the CNN model, and the audio recordings of 30 species of birds were classified. The accuracy of 92.5% and F1-Score of 94.7% were obtained. Chandu et al. [93] reconstructed an ideal dataset using various pre-processing techniques for bird audio recordings. Using each reconstructed audio segment to generate spectrogram as input, the CNN model was used to identify 4 species of birds with 91% accuracy. Anand et al. [94] extracted the audio spectrums of 5 species of birds by STFT, and classified them by using the constructed CNN model, achieving 90% accuracy. Gunawan et al. [95] proposed a two-input scalable convolutional neural network based on EfficientNet and trained it using Mel spectrographs and MFCC. The MAP of 99.27% was achieved on the test data. Permana et al. [96] used the CNN model to classify bird sounds under two conditions, to distinguish between the changes in bird sounds under normal conditions and panic conditions. The classification accuracy on 2 species of bird

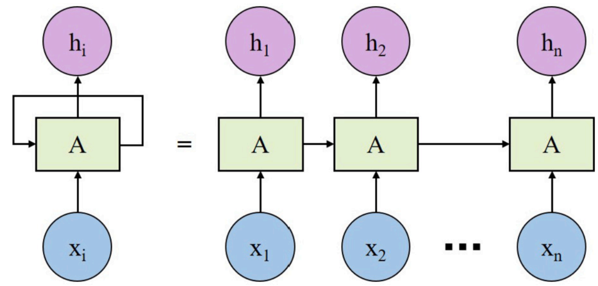

Fig. 10. The structure of RNN [75].

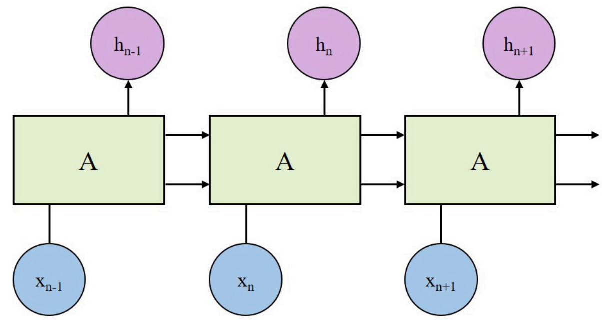

Fig. 11. The structure of LSTM [76].

sounds reached 96.45%, which provided a preliminary study for biological early warning based on bird sounds. Höchst et al. [97] proposed an edge AI system (Bird@Edge) based on the EfficientNet-B3 architecture for recognizing recordings of birds. On soundscape recordings in the Marburg Open Forest, the MAP of up to 95.2% was achieved. Liu et al. [98] proposed an ensemble multi-scale convolutional neural network (EMSCNN) classification framework based on the feature spectrum of birdsong wavelet transform. In the identification experiment of 30 bird species, the accuracy of EMSCNN was 91.49%. Cole et al. [99] evaluated the performance of BirdNET through 13 breeding bird species collected in northwestern California. The results found BirdNET had the average recall of 33% and MAP of 81%. Bota et al. [100] also evaluated the effectiveness of BirdNET. On two mysterious forest bird datasets, Peripatus ater and Certhia brachydactyla, the MAP of BirdNET was 92.6% and 87.8%, respectively. Wang et al. [37] designed Phylogenetic Perspective Neural Networks (PPNN) using hierarchical semantic embedding frameworks and attention mechanisms. A 3D spectrogram was also generated to characterize bird sounds. On the Lishui-Zhejiang Birdsdata (100 species), BirdCLEF2018-Small (150 species), and BirdCLEF2018-Large (500 species) datasets, the classification accuracy was 90.45%, 91.883% and 89.950%, respectively. Swaminathan et al. [101] fine-tuned Wav2vec using backpropagation techniques and effectively used it to identify multiple birds from live recordings through transfer learning techniques. On the Xeno-Canto dataset, F1-Score of 89% was achieved. Lavner et al. [102] proposed an automated framework based on the BirdNET-Analyzer deep learning model to monitor large numbers of birds by vocalizing over long periods. For the 23 bird species that were actually recorded, although the recall was relatively low for a few species, the accuracy was consistently higher than 70% for most species.

4.2.3. *Temporal neural network (TNN)*

Temporal Neural Network (TNN) is a kind of network that takes time series data as input. The most common are Recurrent Neural Network (RNN) [75] and Long Short-Term Memory Network (LSTM) [76]. RNN is a network containing loops to pass information from one time step to the next, and its structure RNN is shown in Fig. 10.

In Fig. 10, after the loop expansion of RNN, it is actually to connect multiple copies of the same network to transmit timing information. The network structure used for replication consists of input values x_1, x_2, …, x_n, cycle unit A and system state values h_1, h_2, …, h_n.

LSTM is an extension of RNN. The network architecture is generally similar to RNN, and its structure is shown in Fig. 11.

Different from RNN, LSTM has added the gating algorithm. The cycle unit of the LSTM contains three gates: input gate, forget gate, and output gate. Compared with the recursive calculation of the system state established by the RNN, the three gates establish a self-loop on the internal state of the loop unit of the LSTM, which can learn long-distance dependencies more effectively.

With continuous adjustment and development, RNN and LSTM are gradually applied to bird sounds recognition research.

Trinh et al. [103] proposed a bird detection method combining CNN and LSTM. They used CNN to extract rich features and LSTM to analyze continuous changes in bird appearance. In the judgment experiments with and without birds, the method achieved 83% MAP, better than CNN alone. Emreakr et al. [104] proposed an automatic bird audio detection method combining CNN and RNN. First, the convolutional layer of CNN was used to extract high-dimensional features, and then the cyclic layer of RNN was used to capture the long-term dependency between features. The results showed that the accuracy of the proposed method was 95.5% on the seen evaluation data of bird sounds. Zhang et al. [105] improved the traditional 3D convolutional structure by using RNN instead of the fully connected layer as the classifier, to further extract the time dimension of the feature graph. By testing the dataset of 4 bird species collected online, MAP achieved 97% and improved classification performance. Solomes and Stowell [22] proposed convolutional recurrent neural network (CRNN) and used C++ to export the model to Bela Mini for embedded system development. The system achieved high-quality bird sounds recognition, and achieved an accuracy of 82.5% on the test set of the International Data Challenge. Xie et al. [106] reconsidered the association between convolutional architecture and cyclic network. They regarded CNN model as the starting point of sequence modeling, and then was combined with LSTM. Meanwhile, the multi-feature fusion method was adopted, and the fusion of LM, MFCC and Chroma based on VGG16 was used as the input of the combined model of CNN and LSTM. On a database of 35 bird species, MAP of the proposed method achieved 97.9%, an improvement of 30.7% over the single-feature model. Gupta et al. [38] combined the CNN with the RNN to obtain a hybrid model to learn the spectrogram features of bird sounds. The best-performing hybrid model achieved the MAP of 67% across 100 different bird species, with the highest accuracy of 90% for single species. Wang et al. [107] proposed the LSTM with coordinate attention. In the Xeno Canto database of 264 bird species, the MAP of the method reached 77.43%, indicating that acoustic features can be used for automatic identification of a large number of bird species. Noumida and Rajan [108] proposed an attention-based layered bidirectional gated cycle unit (BiGRU) model and classified acoustic signals from birds via MFCC. On the Xeno-Canto dataset, F1-Scored of 84% was achieved.

4.3. *Discussion of neural network technology*

4.3.1. *Comparative analysis of different models*

There are significant differences in structure and performance between shallow network models (ANN) and deep network models (such as DNN, CNN, TNN). Shallow models are relatively simple, easy to understand and implement, but may have limited capacity for processing complex sound data. In contrast, deep models can capture more complex feature representations, albeit requiring more data and computational resources. In order to effectively observe the performance of these models, this paper compares and summarizes the mechanism, advantages and disadvantages of these models in more detail, as shown in Table 4.

All layers of neurons in ANN are fully connected, which can be used for nonlinear mapping and global optimization. However, because ANN is too simple, its generalization ability is poor, and the processing effect

Table 4
The performance comparisons of different neural network models, including Artificial Neural Network (ANN), Deep Neural Network (DNN), Convolutional Neural Networks (CNN), Recurrent Neural Network (RNN) and Long Short-Term Memory networks (LSTM).

Models	Mechanism	Advantages	Disadvantages
ANN	All layers of neurons are fully connected.	Nonlinear mapping, global optimization.	Poor generalization ability and poor multidimensional data processing effect.
DNN	The hidden layer is greatly increased, and all neurons are fully connected.	Spatial information is broken and multidimensional data is learned globally.	Local information is lost and the amount of computation expands.
CNN	Convolution kernel extracts features, and downsampling retains information.	Local connection, weight sharing.	The required data is large and the input size is fixed.
RNN	All nodes (loop units) are connected in a chain.	The input size is flexible, and the time sequence extraction ability is strong.	Poor ability to handle data on long-term dependencies.
LSTM	Three gating Settings, two status Settings.	Long-term memory is strong, and sequence modeling is advantageous.	Poor parallel processing capability.

of multi-dimensional data is not good. DNN deepens the network depth by adding hidden layers and fully connects all neurons, breaking spatial information and conducting global learning on multi-dimensional data, but it also leads to local information loss, resulting in computational expansion. CNN uses convolution kernel to extract features and subsampling to retain spatial information, which realizes local connection and weight sharing, but there are also problems such as large amount of required data and fixed input size. RNN connects all nodes (cyclic units) in a single direction according to the chain, and has a strong ability to extract sequential features, but it has poor effect on data processing with long-term dependence. LSTM is an extension of RNN, adding three kinds of gating settings and two kinds of state settings, with strong long-term memory ability, and more advantages in sequence modeling, but poor parallel processing ability.

In practical applications, neural network models typically require input data in a certain format. For audio data, this often involves converting the audio signal into particular feature representations, such as MFCC or other spectral features. Automatic segmentation as a preprocessing step typically aims to extract meaningful sound units, such as syllables or phrases of bird sounds, from raw audio. If the designed neural network model is intended to process entire audio segments or already incorporates some form of sound unit, additional automatic segmentation may not be required. However, if the model input necessitates specific lengths or uniform formats of sound units, or if the raw audio contains multiple sound sources, automatic segmentation may still be a necessary preprocessing step.

According to the model performance evaluation in studies, different neural network architectures and parameter configurations significantly impact accuracy in bird sounds recognition. Future research may focus on enhancing model generalization, optimizing network structure, reducing computational resource consumption, and improving real-time recognition performance. In addition, combining with traditional methods, transfer learning, exploring new data representations, and enhancing valid information are viable directions for improvement.

4.3.2. Citation analysis of different models

Many researchers have used neural network techniques to train their bird sounds recognition models, and these cited studies are summarized in Table 5.

Table 5 provides a summary of the research use of neural network technology for bird sounds recognition, including descriptive information, number of bird species, acoustic features used, classification methods, and performance results. Although ANN may not be as powerful as the deep models, it still excelled in specific applications, achieving an accuracy rate of 93.93%. DNN enhanced abstraction and dimensionality reduction capabilities significantly by increasing the number of hidden layers, with accuracy rates over 73% on large-scale datasets. CNN could effectively extract the spatial features of sound signals through convolution layer and pooling layer, and obtain 55.5%-99% MAPs under different prediction conditions, which highlights its effectiveness in sound recognition. TNN, particularly RNN and LSTM, achieved a MAP of 83% in bird sound time series experiments, demonstrating their advantage in analyzing continuous variations.

Similar to Fig. 4, this paper also presents pie charts depicting the percentage of categories of neural network technology (Left in Fig. 12) and categories of feature used (Right in Fig. 12), respectively. In the categories of neural network technology, SNM (primarily ANN and SNN), DNN, CNN, TNN (predominantly RNN and LSTM) and Mix accounted for 7%, 14%, 50%, 18%, 11%, respectively. CNN, which accounts for almost half of all use, is the most prevalent neural network technology, proving its effectiveness on the bird sounds recognition tasks. Conversely, SNM exhibits the lowest usage rate potentially due to structural limitations. In the categories of feature used, the proportion from most to least is Spectrogram (39%), MFCC (17%), Hybrid feature (11%), Other (11%), Frequency (9%), Transformation (7%), Energy sequence (6%). With the enhancement of the information extraction capabilities of neural network models, the usage rate of spectrogram, carrying more information elements, has gradually surpassed that of MFCC, becoming the most commonly employed acoustic feature in bird sounds recognition. Concurrently, the application of hybrid feature is also increasing, ranking just below MFCC and spectrogram, and has become a common method to improve the recognition performance.

Analyzing these references, selecting appropriate neural network models and optimization methods can effectively promote the efficiency of bird sounds recognition. Transfer learning, a commonly used method for speech recognition tasks, uses pre-trained networks to reduce training data and training time. Model fusion is also a commonly used method, which is applicable not only between neural network models, but also between neural network models and traditional models. Inspired by an interdisciplinary approach, the attention mechanisms offer new perspectives for bird sounds recognition. In the follow-up research, the CNN model was mainly used as the modeling basis, and a series of methods such as attention mechanism were combined, to conduct improvement on the bird sounds recognition experiment [13,37,101,108], including the experimental exploration from our previously published work.

5. Conclusion

This paper systematically reviews the advancements in bird sounds recognition technologies based on acoustic monitoring technology. In view of the unsolved and valuable problems, some suggestions for future research are put forward: 1) diversity and quality of datasets. Despite the availability of multiple avian sound datasets, variations exist in species coverage, sample size, and recording quality. Further development of more comprehensive, diverse, and accurately labeled datasets is necessary to support broader research on bird sounds recognition. This involves researching automated or semi-automated labeling techniques, generative models, and semi-supervised or unsupervised learning methods. 2) Data fusion. Most studies have focused on a single acoustic signal

Table 5
The citation details of neural network models in bird sound recognition.

Year	Study	Number of classes	Feature use	Classifier	Performance
2016	Raghuram et al. [70]	33	Fast Fourier Transform features	ANN	Accuracy: 93.93%
2020	Mohanty et al. [71]	14	Permutation pair frequency matrix	SNN	Accuracy: 92%
2021	Ghani and Hallerberg [11]	10	Spectral centroid, spectral roll-off, zero-crossing rate, spectral bandwidth, root-mean-square energy, and MFCC	SNN	MAP: 86%-94% (mean accuracy probability)
2014	Koops et al. [78]	501	MFCC	DNN	Accuracy: 73%
2016	Boulmaiz et al. [79]	36	Gammatone teager energy cepstral coefficients (GTECC)	DNN	EER: 18.47% (Equal error rates)
2017	Fazeka et al. [80]	1500	Audio and metadata	DNN	MAP: 51.1%-57.9%
2017	Chakraborty et al. [81]	26	MFCC	DNN	Accuracy: 63.1%-91.5%
2019	Jancovi et al. [82]	48	A sequence of the frequency and normalized magnitude values of the sinusoid	DNN+HMM	Accuracy: 95.4%–98.7%
2021	Kahl et al. [27]	948	High temporal resolution of input spectrograms	DNN	MAP: 79.1%
2016	Sprengel et al. [85]	999	Spectrogram	CNN	MAP: 55.5%-68.8%
2016	Tóth and Czeba [86]	999	Spectrogram	AlexNet	MAP: 33%-40%
2017	Kahl et al. [87]	1500	Spectrogram	CNN+FC	MAP: 60.5%-68.7%
2017	Fritzler et al. [88]	1500	Spectrogram	Inception-v3	MAP: 49.6%-56.7%
2018	Sankupellay et al. [89]	46	Spectrogram	ResNet50	Accuracy: 60%-72%
2018	Stowell and Konovalov [90]	2	MFCC	CNN	Accuracy: 88%
2018	Incze et al. [91]	1500	Spectrogram	CNN	Accuracy: 70.9%
2019	Kucuktopcu et al. [14]	21	Spectrogram	CNN	Accuracy: 90%
2019	Ruff et al. [25]	7	Spectrogram	DCNN	Accuracy: 91.5%
2019	Xie et al. [26]	43	Time-frequency representations	CNN	MAP: 86.31%, F1-score: 93.31%
2020	Tubaro and Mindlin [92]	6	Spectrogram	CNN	Accuracy: 89%, Recall: 84%, F1-Score: 84%
2020	LeBien et al. [24]	24	Mel-spectrogram	CNN	MAP: 89.3%
2020	Kiskin et al. [28]	30	Wavelet transformations	CNN	Accuracy: 92.5%, F1-score: 94.7%
2020	Chandu et al. [93]	4	Spectrogram	CNN	Accuracy: 91%
2021	Anand et al. [94]	5	Spectrogram	CNN	Accuracy: 90%
2021	Permana et al. [96]	2	Spectrogram	CNN	Accuracy: 96.45%
2021	Gunawan et al. [95]	–	Mel-spectrogram and MFCC	CNN	MAP: 99.27%
2022	Höchst et al. [97]	–	Spectrogram	CNN	MAP: 95.2%
2022	Liu et al. [98]	30	Spectrogram from the wavelet transform	CNN	Accuracy: 91.49%
2022	Cole et al. [99]	13	–	CNN	MAP: 81%, Average recall: 33%
2023	Bota et al. [100]	2	–	CNN	MAP: 92.6%, 87.8%
2024	Wang et al. [37]	100, 150, 500	3D spectrogram	CNN	Accuracy: 90.45%, 91.883%, 89.950%
2024	Swaminathan et al. [101]	–	MFCC	1D-CNN++	F1-Score = 89%
2024	Lavner et al. [102]	23	Vocalogram	CNN	Accuracy > 70%
2016	Trinh et al. [103]	2	Downsampled timing vector	CNN+LSTM	MAP: 83%
2017	Emreakr et al. [104]	2	Dominant frequency and log mel-band energy	CNN+RNN	Accuracy: 95.5%
2019	Zhang et al. [105]	4	Mel-spectrogram	3D-CNN+RNN	MAP: 97%
2020	Solomes and Stowell [22]	2	Spectrogram	CRNN	Accuracy: 82.5%
2020	Xie et al. [106]	35	LM, MFCC and Chroma	CNN+LSTM	MAP: 97.9%
2021	Gupta et al. [38]	100	Mel-spectrogram	RNN	MAP: 67% Accuracy: 90% (Highest)
2022	Wang et al. [107]	264	Mel-Spectrogram and MFCC	LSTM	MAP: 77.43%
2022	Noumida and Rajan [108]	–	MFCC	RNN	F1-Score: 84%

and have made use of limited data. Further exploration on data fusion methods is needed to enhance the accuracy of bird sounds recognition technologies. This includes the multimodal fusion methods combining audio, visual and more, the multi-feature fusion methods integrating signal features, spectrogram features and more, and the interdisciplinary fusion methods incorporating biological, ecological, data science and more. 3) Generalization capability of recognition models. Existing models may perform well on specific datasets, but lack robustness when applied to new or different bird sound data. Learning methods such as transfer learning or multi-task learning can be studied to improve the applicability of the model, and intensifying researches for practical applications in biodiversity assessment and habitat monitoring are also essential. 4) Optimization of recognition models. Despite a large number of existing recognition models, there is ample room for optimization in structure and performance. Traditional models require enhancements to their adaptability and robustness to better process complex bird sound data. Deep learning models should focus on developing architectures with more computing power, more interpretability and fewer parameters, so as to reduce the computational burden and improve the training efficiency. Further combined research on traditional models, deep learning models or emerging technologies could leverage their collective strengths more effectively.

CRediT authorship contribution statement

Daidai Liu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Hanguang Xiao:** Writing – review & editing, Supervision, Funding acquisition. **Kai Chen:** Visualization, Data curation.

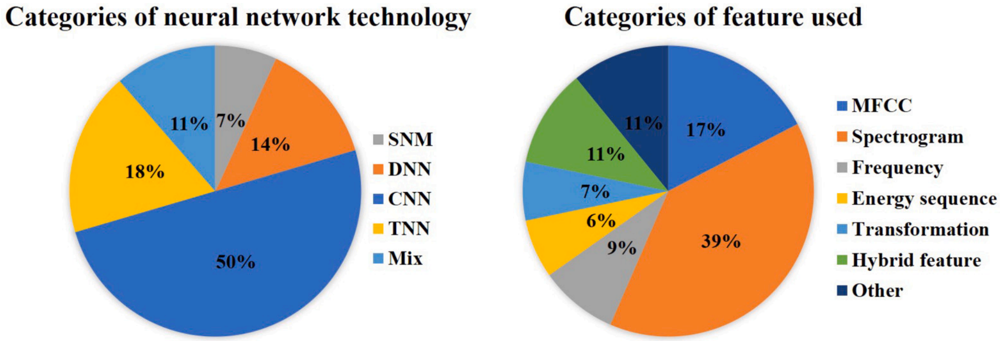

Fig. 12. Percentage pie chart of categories of neural network technology and feature used. SNM and TNN are not specific network names, but categorizations of some networks. Frequency, Energy sequence, Transformation are not specific feature names, but categorizations of some features.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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